



## **Objective:**

This project focuses on organizing, analyzing, and extracting insights from a product data set. Using fundamental Excel functions, the workflow transforms raw product metrics such as pricing, Price range, and product count into a structured summary that highlights inventory health and product performance.

## **Data set Description:**

This data set contains detailed retail information for a variety of Electronic products. It tracks core product attributes, pricing structures, and current stock volumes. The data is used to analyze inventory levels, categorize items and monitor overall inventory health.

### **Key Data Included:**

Product Details: Basic identification including names and categories.

Pricing Metrics: Individual product price

Inventory: Individual product quantity.

## **Column Description:**

| Column Name  | Description                                              |
|--------------|----------------------------------------------------------|
| Product ID   | ID of the product                                        |
| Product Name | Name of the product                                      |
| Brand Name   | It shows brand name of the product                       |
| Price        | It shows price of the product                            |
| Quantity     | It shows the quantity of product we have.                |
| Category     | It shows the product is basically from which categories. |

## Data Exploration:

We have done the data exploration using excel formula's as per the requirements.

The Requirements and Outcomes are,

### 1. Basic Aggregation

- Calculate the total price of all products in the dataset(SUM())
- Determine how many products exist in the dataset(COUNTA())
- Calculate the average price of the products(AVERAGE())
- Identify the minimum and maximum price among all products.(MIN() & MAX()).

### Outcomes:

| 1. SUM, COUNT and AVERAGE |                                     |                         |            |
|---------------------------|-------------------------------------|-------------------------|------------|
| Function                  | Requirements                        | Function In             | Result     |
| SUM()                     | Total price of all products         | SUM(\$E\$2:\$E\$35)     | ₹10,100.00 |
| COUNT()                   | How many products exist in data set | COUNTA(\$C\$2:\$C\$35)  | 34         |
| AVERAGE()                 | Average price of the product        | AVERAGE(\$E\$2:\$E\$35) | 297        |

  

| 2. Minimum and Maximum Values |                                     |                     |        |
|-------------------------------|-------------------------------------|---------------------|--------|
| Function                      | Requirements                        | Function In         | Result |
| MIN()                         | Minimum price among all the product | MIN(\$E\$2:\$E\$35) | ₹30    |
| MAX()                         | Maximum price among all the product | MAX(\$E\$2:\$E\$35) | ₹1,000 |

### 2. Logical Function

- Use the IF function to categorize products,  
If Price  $\geq$  \$500 → classify as High Price  
If Price  $<$  \$500 → classify as Standard Price

### Outcomes:

| 3. Logical Function - IF |                                                |                                                                         |                       |
|--------------------------|------------------------------------------------|-------------------------------------------------------------------------|-----------------------|
| Function                 | Requirements                                   | Function In                                                             | Result                |
| IF()                     | Categorized product Based on the Price.        |                                                                         |                       |
|                          | Price $\geq$ 500 Classify as <b>High Price</b> | IF(E2 $\geq$ 500,"High Price",IF(E2<500,"Standard Price","High Price")) | <b>High Price</b>     |
|                          | Price <500 Classify as <b>Standard Price</b>   | IF(E2 $\geq$ 500,"High Price",IF(E2<500,"Standard Price","High Price")) | <b>Standard Price</b> |

### 3. Conditional Function:

- calculate the total price of products in the Electronics category (SUMIF)
- Determine the number of products with a price less than \$100 (COUNTIF)

## Outcomes

| 4.Conditional Functions - SUMIF and COUNTIF |                                                        |                                                    |        |
|---------------------------------------------|--------------------------------------------------------|----------------------------------------------------|--------|
| Function                                    | Requirements                                           | Function In                                        | Result |
| SUMIF()                                     | Total price of products in <b>Electronics</b> category | SUMIF(\$G\$2:\$G\$35,"Electronics",\$E\$2:\$E\$35) | ₹8,050 |
| COUNTIF()                                   | No.of Product with the price < 100                     | COUNTIF(E2:E35,"<100")                             | 11     |

## 4. Text Formatting:

- Create a new column named Day with the first 2 characters of each 'Product ID' using the LEFT function.
- Create a new column named Country Code by extracting the last 2 characters from the 'Product ID' column using the RIGHT function.
- Create a new column named Month by extracting the 4th to 6<sup>th</sup> characters from the 'Product ID' column using the MID function.

## Outcomes:

| 5. Text Formatting - LEFT, RIGHT, MID |                                                                                                  |             |        |
|---------------------------------------|--------------------------------------------------------------------------------------------------|-------------|--------|
| Function                              | Requirements                                                                                     | Function In | Result |
| LEFT()                                | Create a new column named Day with the first 2 char of each Product ID                           | LEFT(B2,2)  | 28     |
| RIGHT()                               | Create a new column named Country Code by extracting the last 2 characters from the 'Product ID' | RIGHT(B2,2) | US     |
| MID()                                 | Create a new column named Month by extracting the 4th to 6th characters from the                 | MID(B2,4,3) | JAN    |

## CONCLUSION:

By applying fundamental Excel formulas to the raw dataset, the project requirements were successfully met. The formulas successfully converted chaotic product metrics into a structured, actionable summary of inventory health and performance. The Overall outcomes are,

| Product   | D  | Country | Month | Product Name    | Brand Name | Price | Quantity | Category    | Price Range    |
|-----------|----|---------|-------|-----------------|------------|-------|----------|-------------|----------------|
| 28-JAN-US | 28 | US      | JAN   | Laptop          | Dell       | 1000  | 30       | Electronics | High Price     |
| 15-FEB-US | 15 | US      | FEB   | Sneakers        | Nike       | 80    | 15       | Fashion     | Standard Price |
| 03-MAR-US | 03 | US      | MAR   | Coffee Maker    | Keurig     | 150   | 40       | Kitchen     | Standard Price |
| 11-APR-US | 11 | US      | APR   | Smartphone      | Samsung    | 900   | 25       | Electronics | High Price     |
| 22-MAY-US | 22 | US      | MAY   | Backpack        | North Face | 70    | 20       | Outdoor     | Standard Price |
| 07-JUN-UK | 07 | UK      | JUN   | Headphones      | Sony       | 200   | 45       | Electronics | Standard Price |
| 19-JUL-UK | 19 | UK      | JUL   | T-shirt         | Adidas     | 50    | 5        | Fashion     | Standard Price |
| 25-AUG-UK | 25 | UK      | AUG   | Blender         | Ninja      | 90    | 35       | Kitchen     | Standard Price |
| 05-SEP-UK | 05 | UK      | SEP   | Tablet          | Apple      | 500   | 50       | Electronics | High Price     |
| 14-OCT-UK | 14 | UK      | OCT   | Hiking Boots    | Timberland | 150   | 10       | Outdoor     | Standard Price |
| 17-JUN-IN | 17 | IN      | JUN   | Laptop          | HP         | 950   | 25       | Electronics | High Price     |
| 25-NOV-AU | 25 | AU      | NOV   | Sneakers        | Adidas     | 90    | 40       | Fashion     | Standard Price |
| 08-DEC-DE | 08 | DE      | DEC   | Coffee Maker    | Nespresso  | 120   | 35       | Kitchen     | Standard Price |
| 18-FEB-CA | 18 | CA      | FEB   | Smartwatch      | Fitbit     | 150   | 15       | Electronics | Standard Price |
| 16-APR-ES | 16 | ES      | APR   | Headphones      | Beats      | 250   | 20       | Electronics | Standard Price |
| 21-AUG-CA | 21 | CA      | AUG   | Laptop Bag      | Samsonite  | 50    | 35       | Accessories | Standard Price |
| 20-AUG-CN | 20 | CN      | AUG   | Smartwatch      | Huawei     | 160   | 15       | Electronics | Standard Price |
| 27-JAN-JT | 27 | JT      | JAN   | Laptop          | Asus       | 980   | 10       | Electronics | High Price     |
| 01-MAR-UK | 01 | UK      | MAR   | Sunglasses      | Oakley     | 150   | 15       | Fashion     | Standard Price |
| 14-AUG-US | 14 | US      | AUG   | Camping Tent    | Coleman    | 200   | 10       | Outdoor     | Standard Price |
| 14-MAY-RU | 14 | RU      | MAY   | Camera          | Sony       | 700   | 50       | Electronics | High Price     |
| 09-JAN-CA | 09 | CA      | JAN   | Microwave       | Panasonic  | 80    | 20       | Kitchen     | Standard Price |
| 19-JUL-BR | 19 | BR      | JUL   | Fitness Tracker | Xiaomi     | 150   | 30       | Electronics | Standard Price |
| 21-AUG-CA | 21 | CA      | AUG   | Laptop Bag      | Samsonite  | 50    | 35       | Accessories | Standard Price |
| 20-SEP-CA | 20 | CA      | SEP   | Smartphone      | Google     | 800   | 45       | Electronics | High Price     |
| 03-JUN-CA | 03 | CA      | JUN   | Sunglasses      | Ray-Ban    | 150   | 25       | Fashion     | Standard Price |
| 11-JUL-CA | 11 | CA      | JUL   | Blender         | Vitamix    | 400   | 40       | Kitchen     | Standard Price |
| 16-APR-ES | 16 | ES      | APR   | Headphones      | Beats      | 250   | 20       | Electronics | Standard Price |
| 07-MAR-CA | 07 | CA      | MAR   | Dress           | Zara       | 60    | 30       | Fashion     | Standard Price |
| 13-APR-CA | 13 | CA      | APR   | Toaster         | Hamilton   | 40    | 10       | Kitchen     | Standard Price |
| 24-MAY-CA | 24 | CA      | MAY   | Fitness Tracker | Garmin     | 150   | 5        | Electronics | Standard Price |
| 02-DEC-CA | 02 | CA      | DEC   | Jeans           | Levi's     | 50    | 50       | Fashion     | Standard Price |
| 17-JUN-IN | 17 | IN      | JUN   | Laptop          | HP         | 950   | 25       | Electronics | High Price     |
| 09-JUL-FR | 09 | FR      | JUL   | Watch           | Casio      | 100   | 20       | Accessories | Standard Price |

| Data Exploration - Module 1 - Excel Task     |                                                                                                  |                                                                   |                |
|----------------------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|----------------|
| 1. SUM, COUNT and AVERAGE                    |                                                                                                  |                                                                   |                |
| Function                                     | Requirements                                                                                     | Function In                                                       | Result         |
| SUM()                                        | Total price of all products                                                                      | SUM(\$E\$2:\$E\$35)                                               | ₹10,100.00     |
| COUNT()                                      | How many products exit in data set                                                               | COUNTA(\$C\$2:\$C\$35)                                            | 34             |
| AVERAGE()                                    | Average price of the product                                                                     | AVERAGE(\$E\$2:\$E\$35)                                           | 297            |
| 2. Minimum and Maximum Values                |                                                                                                  |                                                                   |                |
| Function                                     | Requirements                                                                                     | Function In                                                       | Result         |
| MIN()                                        | Minimum price among all the product                                                              | MIN(\$E\$2:\$E\$35)                                               | ₹30            |
| MAX()                                        | Maximum price among all the product                                                              | MAX(\$E\$2:\$E\$35)                                               | ₹1,000         |
| 3. Logical Function - IF                     |                                                                                                  |                                                                   |                |
| Function                                     | Requirements                                                                                     | Function In                                                       | Result         |
| IF()                                         | Categorized product Based on the Price.                                                          |                                                                   |                |
|                                              | Price >= 500 Classify as <b>High Price</b>                                                       | IF(E2>=500,"High Price",IF(E2<500,"Standard Price","High Price")) | High Price     |
|                                              | Price <500 Classify as <b>Standard Price</b>                                                     | IF(E2>=500,"High Price",IF(E2<500,"Standard Price","High Price")) | Standard Price |
| 4. Conditional Functions - SUMIF and COUNTIF |                                                                                                  |                                                                   |                |
| Function                                     | Requirements                                                                                     | Function In                                                       | Result         |
| SUMIF()                                      | Total price of products in <b>Electronics</b> category                                           | SUMIF(\$G\$2:\$G\$35,"Electronics",\$E\$2:\$E\$35)                | ₹8,050         |
| COUNTIF()                                    | No of Product with the price < 100                                                               | COUNTIF(E2:E35,"<100")                                            | 11             |
| 5. Text Formatting - LEFT, RIGHT, MID        |                                                                                                  |                                                                   |                |
| Function                                     | Requirements                                                                                     | Function In                                                       | Result         |
| LEFT()                                       | Create a new column named Day with the first 2 char of each Product ID                           | LEFT(B2,2)                                                        | 28             |
| RIGHT()                                      | Create a new column named Country Code by extracting the last 2 characters from the 'Product ID' | RIGHT(B2,2)                                                       | US             |
| MID()                                        | Create a new column named Month by extracting the 4th to 6th characters from the 'Product ID'    | MID(B2,4,3)                                                       | JAN            |